Activity: quasi-Poisson inference

t-tests for quasi-Poisson models

For a quasi-Poisson model,

$$\hat{\beta} \approx N(\beta, \phi(\mathbf{X}^T\mathbf{V}\mathbf{X})^{-1})$$

where $\mathbf{V} = \text{diag}(V(\hat{\mu}_i)) = \text{diag}(\hat{\mu}_i)$.

To test $H_0 : \beta_j = b$ for one of the coefficients β_j, the test statistic

$$Z = \frac{\hat{\beta}_j - b}{\sqrt{\phi[(\mathbf{X}^T\mathbf{V}\mathbf{X})^{-1}]_{jj}}} \approx N(0,1)$$

where $[(\mathbf{X}^T\mathbf{V}\mathbf{X})^{-1}]_{jj}$ denotes the diagonal entry of $(\mathbf{X}^T\mathbf{V}\mathbf{X})^{-1}$ corresponding to β_j.

However, ϕ is typically unknown, in which case we use the test statistic

$$t = \frac{\hat{\beta}_j - b}{\sqrt{\hat{\phi}[(\mathbf{X}^T\mathbf{V}\mathbf{X})^{-1}]_{jj}}}$$

Facts

Recall the following:

- If $Z \sim N(0,1)$ and $X \sim \chi^2_d$ are independent, then $\dfrac{Z}{\sqrt{X/d}} \sim t_d$
- Common estimates of ϕ include

$$\hat{\phi}_D = \frac{D(\mathbf{y}, \hat{\mu})}{n-p} \qquad \hat{\phi}_P = \frac{1}{n-p}\sum_{i=1}^{n}\frac{(Y_i - \hat{\mu}_i)^2}{V(\hat{\mu}_i)}$$

- We have argued that

$$\frac{D(\mathbf{y}, \hat{\mu})}{\phi} \approx \chi^2_{n-p} \qquad \frac{1}{\phi}\sum_{i=1}^{n}\frac{(Y_i - \hat{\mu}_i)^2}{V(\hat{\mu}_i)} \approx \chi^2_{n-p}$$

- If $X_1 \sim \chi^2_{d_1}$ and $X_2 \sim \chi^2_{d_2}$ are independent, then

$$\frac{X_1/d_1}{X_2/d_2} \sim F_{d_1,d_2}$$

- If we have two nested models (full and reduced), then

$$\frac{D(\mathbf{y}, \hat{\mu}_{reduced}) - D(\mathbf{y}, \hat{\mu}_{full})}{\phi} \approx \chi^2_{p_{full}-p_{reduced}}$$

Questions

Show the following. You can assume the necessary independence.

1.

$$t = \frac{\hat{\beta}_j - b}{\sqrt{\hat{\phi}[(\mathbf{X}^T\mathbf{V}\mathbf{X})^{-1}]_{jj}}} \approx t_{n-p}$$

2.

$$\frac{(D(\mathbf{y}, \hat{\mu}_{reduced}) - D(\mathbf{y}, \hat{\mu}_{full}))/(p_{full} - p_{reduced})}{\hat{\phi}} \approx F_{p_{full}-p_{reduced},\ n-p_{full}}$$